SLANT carrier board: design record

Carthage College Space Sciences, 4 November 2026.

The carrier board connects the Raspberry Pi Compute Module 5 (CM5) to the host spacecraft and to the XIMEA camera head. It isolates the host's unregulated bus, makes and supervises the 5 V rails, and gives the CM5 its host interfaces and housekeeping measurements. This record gives the design standard, the isolation scheme, the reasoning and arithmetic behind each choice, the derating, the parts and the pin maps. The schematic, parts list, layout and KiCad project are on the Electronics tab. XIMEA develops the flight board from this design.

Status

- **Schematic.** 172 parts and 109 nets on seven sheets. Every symbol pin is taken from the vendor's datasheet, and every passive carries its value and the basis for it. KiCad 10.0.7 reports 0 errors and 0 warnings in the electrical rules check, and the exported netlist matches the design data net for net.
- **Layout.** Four-layer polyimide board, 2.4 mm thick, designed to IPC-6012 Class 3. Every part is placed. The bus power chain, the USB, RS-422 and CAN pairs and the shunt sense lines are routed. KiCad's design-rule check finds 0 errors, and an independent check of the copper geometry finds no violation of the isolation barrier or the chassis spacing on any layer.

Fabrication standard and stackup

- IPC-6012 Class 3 for the bench and the flight boards, designed to IPC-2221 Class 3 with IPC-2222. Goddard's GSFC-STD-8001 asks for Class 3/A, the space addendum IPC-6012FS, which a NASA customer may require.
- Polyimide laminate, four copper layers, 2.4 mm, ENIG finish. The assembled board is conformal coated.

Layer	Use	Thickness
F.Mask		0.01
F.Cu	Components, signals, power pours	0.035 (1 oz)
Prepreg	Polyimide	0.15
In1.Cu	GND plane, PRI_RTN island under the host-side zone	0.035
Core	Polyimide	1.94
In2.Cu	GND plane and PRI_RTN island, as In1	0.035
Prepreg	Polyimide	0.15
B.Cu	CM5 connectors, input filter, LDO, signals	0.035 (1 oz)
B.Mask		0.01

Dielectric constant 4.2 and loss tangent 0.02 are assumed until the fabricator names its polyimide laminate.

Net classes.

Class	Nets	Track	Pair width / gap	Clearance	Basis
USB90	HEAD_SS*, HEAD_USB2_, *MAINT_USB_*	0.205	0.205 / 0.15	0.2	90 ohm differential (CM5 datasheet); IPC-2141 estimate over 0.15 mm, er 4.2: 89 ohm; 85-96 ohm over er 3.8-4.6 and prepreg 0.13-0.17 mm. The fab sets the final widths with impedance coupons.
Diff120	CMD_IN_, *TLM_*, BULKOUT_, *BULKIN_*, PPS_IN_*, CANH, CANL	0.15	0.15 / 0.30	0.2	120 ohm, the termination value; cable impedance to confirm
BusSide	BUS_*, FILT_DAMP, PRI_RTN	1.0		0.2	up to 2.5 A (limiter setting)
BusSideSignal	LIM_*	0.25		0.2	
Host	HOST_THERM_*	0.25		0.2	
Chassis	CHASSIS	1.0		0.2	
Power	5V_BUS, SW1_OUT, SW2_OUT, 5V_CM5, 5V_HEAD, GND	1.0		0.2	up to 2.2 A (SW1 limit)
PowerLow	3V7_A, VREF, 5V_RS422, 5V_BIAS, CM5_3V3	0.5		0.15	
Default	everything else	0.2	0.2 / 0.15	0.15	

The 1.0 mm and 0.5 mm power widths come from the IPC-2221 conductor chart for 1 oz outer copper. In vacuum the copper sheds heat only by conduction, and conductors run 35 to 55 % hotter than in air, so these widths are checked against IPC-2152 before release.

Board rules: minimum clearance 0.15 mm, track 0.15 mm, via 0.55 mm with 0.3 mm hole, annular ring 0.125 mm, hole to hole 0.25 mm, copper to edge 0.5 mm. Default via 0.7 mm with a 0.35 mm hole (a 6.9:1 aspect ratio in 2.4 mm).

Custom rules: the host-to-payload barrier (2.5 mm), chassis (0.8 mm), J1's internal domain gaps (1.5 mm and 0.8 mm), the not-fitted chassis tie R14/C19, and J1's latch slots (Harwin's layout, 0.25 mm to the end pads).

Isolation

- The board is designed for a 500 Vdc test between the host side and the payload, and from each to chassis. The CCG10 converter is rated 1,500 Vdc input to output, so the test stays at a third of that rating.

- Three domains: **host side** (the bus input, fuse, limiter and filter nets, PRI_RTN, and the host's thermistor), **payload** (everything on GND) and **chassis**.
- Spacings follow IPC-2221B Table 6-1 for 301 to 500 V: internal conductors 0.25 mm; external uncoated at sea level 2.5 mm; external coated 0.8 mm.
- Host side to payload: 2.5 mm on every layer, so the 500 V test also passes before coating. The host-side parts sit in one zone surrounded by a 2.5 mm moat.
- Chassis to either domain: 0.8 mm, so the chassis test runs after coating.
- Inside the harness header J1 (1.25 mm pitch) the domains are two pitches apart, a 1.6 mm pad gap: 500 V after coating, 300 V before. J1's pads are 0.90 mm to widen the gaps between pins.

Sheets

1. **Root.** The sheet index and the power domains.
2. **Power entry, limiter and isolated converter.** J1, fuse F1 (4 A, TDK's maximum for the CCG10-24) and TVS D5; the TPS26636 latch-off eFuse with its undervoltage, current-limit, monitor and slew networks; TDK's Class A input filter with ceramic capacitors and a damping leg; the CCG10 converter, its trim and the chassis connections.
3. **5 V rails, switches and supervision.** SW1 and SW2 (TPS259470L) with their shared overvoltage divider, shunts and INA293 amplifiers; the always-on 3.71 V rail from the TPS73801; the TL7700 supervisor and the TPS3435 watchdog.
4. **Compute module.** Both 100-pin CM5 connectors, all 200 pins named from CM5 datasheet Table 4.
5. **Host interfaces.** CAN: MCP2518FD controller and SN65HVD233 transceiver, with an ESD2CAN24-Q1 and an optional split termination (not fitted). RS-422: two SN65C1168E, with termination, fail-safe bias, 5 V to 3.3 V dividers and load switch U18.
6. **Housekeeping.** REF5040 reference, ADC128S102 and its eight channels, three thermistors, and the host-read thermistor RT4.
7. **Maintenance port and head link.** J2 (Molex 87831-1420) with two TPD4E05U06, and J3, the flex link to the camera head.

Design decisions

1. **No CM5 pin is driven before the CM5 has power.** The CM5 datasheet (section 3) says "No pins should be powered before the 5 V rail is active".
   - Every part that drives a CM5 pin runs from the CM5's own 3.3 V output (CM5_3V3) or from 5V_RS422, which CM5_3V3 switches on.
   - That covers the CAN controller and transceiver, the ADC's digital supply, and all pull-ups on CM5 lines.
   - The RS-422 receiver outputs (up to 5 V) reach the CM5's 3.3 V inputs through 0.6 dividers.

- The watchdog runs from the always-on 3V7_A. Its WDI pin is an input with no internal pull-up (TPS3435 datasheet), so it does not drive the CM5's GPIO6.

2. **RS-422 supply gated by the CM5's 3.3 V.** The CM5 powers up its 5 V input, then PMIC_EN, then its 3.3 V.
   - Load switch U18 (TPS22919, SC-70-6) feeds the RS-422 parts from 5V_CM5. Its ON pin is CM5_3V3 itself (VIH 1 V).
   - QOD is tied to VOUT, so 5V_RS422 discharges through 24 Ω when CM5_3V3 falls.
   - The fail-safe bias draws from 5V_RS422 through D4, so a host driving the lines while the CM5 is off cannot back-feed the CM5's 5 V pins.
   - With the BAS70W's forward drop of about 0.7 V, the bias across each 120 Ω termination is 0.24 V, above the 0.2 V receiver threshold.
3. **Watchdog strapped to a fixed timeout of 64 s** (57.6 s minimum). TPS3435CAIEGDDFR: 1 s base timeout, multiplier strapped to 64, output pulse 200 ms.
   - Only WDI connects to the CM5. The CM5 must boot and service the watchdog within 57 s. Fitting R28 instead of R27 gives 32 s.
   - VDD is 3.71 V. The WDI high threshold is 0.7 × VDD = 2.60 V; a 3.3 V CM5 GPIO meets it.
4. **The watchdog output drives SW1's enable.** It is wired with the TL7700 reset; both outputs are open drain, pulled up to 3V7_A by 33 kΩ. A watchdog trip removes the CM5's whole 5 V. Cycling EN also clears a latched SW1 fault.
5. **Bus-side limiter in latch-off mode.**
   - The TPS26636 MODE pin is left open, which selects latch-off. The host's per-pass power cycle resets it; on the bench, pulling SHDN (TP2) low and releasing it does the same.
   - Current limit 18 / 7.15 kΩ = 2.52 A, against a maximum input of 1.25 A (10 W at 9 V).
   - UVLO rises at 8.45 V (8.28 to 8.62 V) and falls at 7.90 V.
   - The output clamp (35.7 to 39 V) holds the CCG10's input below 39 V during a bus surge. At the 36 V bus maximum the clamp may begin to regulate; at that voltage the load draws about 0.3 A, so the dissipation stays under 0.1 W.
   - IMON (0.84 V/A) and FLT go to test points only, so the CM5 cannot see a limiter trip. In latch-off mode a trip removes the CM5's power anyway.
6. **Bus voltage and current are not measured on the board.** The limiter sits on the bus side of the converter's isolation. Measuring them here would bridge that isolation, so the host's power switch has to report them.
7. **ADC reference 4.096 V (REF5040, U19).** It runs from 5V_BUS and needs at least 4.30 V.
   - One code is 1.00 mV. The reference contributes 0.05 % initial error and 3 ppm/°C.
   - The thermistor dividers are biased from the same reference, so the thermistor readings stay ratiometric.
   - VA decoupling follows the ADC datasheet (1 µF and 100 nF within 1 cm), plus 10 µF. The total, 11.1 µF of ceramic, lies inside the REF50xx's stable range of 1 to 50 µF with ESR at

most 1.5 Ω.

- The ADC's VD (CM5_3V3, 3.465 V maximum) stays below VA, as the ADC requires.

8. **Always-on 3.71 V rail.** 3V7_A supplies the shunt amplifiers, the watchdog, its strap resistors and the SW1 enable pull-up. Each of these parts accepts 3.71 V: TPS3435 VDD 1.04 to 6 V; TPS25947 EN/UVLO absolute maximum 6.5 V.
9. **SW1 and SW2 (TPS259470L).**
   - Current limits: SW1 1.97 A (1.78 to 2.17 A); SW2 0.89 A (about 0.80 to 0.98 A; TI states ±10 % above 1 A only).
   - SW2's maximum plus the CM5's 0.9 A stays under the CCG10's 2 A, so a head fault trips SW2 before the converter enters hiccup.
   - Overcurrent blanking 3.9 ms (4.7 nF on ITIMER), then latch-off.
   - Output slew 0.91 V/ms (2.2 nF on DVDT): 5 V in 5.5 ms.
   - Both OVLO pins share one divider: they open above 5.69 V and re-close below 5.17 V.
   - FLT outputs go to test points. ADC channel 6 (5V_HEAD / 2) shows SW2's state to the CM5.
10. **Input filter with ceramic capacitors.** TDK's filter uses a 100 µF electrolytic at the input; its ESR damps the LC resonance.
    - The ceramic replacement places the damping explicitly: 1.0 Ω in series with 28 µF across the converter-side capacitor C2 (9.4 µF).
    - L1 and C2 give a characteristic impedance of sqrt(10 µH / 9.4 µF) = 1.03 Ω. With a damping capacitor three times C2, Middlebrook's optimum series resistance is about 0.75 Ω; 1.0 Ω was chosen.
    - The filter's output impedance has to be checked against the CCG10's input impedance on the bench. Class A emissions also need a test.
11. **RT4, a host-read thermistor** (J1 pins 23, 24). The host reads it while the payload is off, so it can check the base plate before powering the payload.
12. **Converter trim and supervisor threshold.**
    - R72 trims the CCG10 up: Vout = 5.02 V + 10.10 / (15 + Rb[kΩ]), 5.05 V at 324 kΩ, selected on test for 5.05 to 5.08 V. Over ±30 °C that gives 5.02 to 5.11 V, below the converter's 5.25 V overvoltage minimum.
    - TL7700C SENSE threshold 490 to 510 mV (−40 to 85 °C). The falling trip is 4.67 V nominal (4.57 to 4.76 V); the release is at most 4.83 V, below the trimmed 5.02 V minimum.
13. **Power between passes is switched by the host.** The host turns SLANT's bus on about a minute before each sunset, at a time SLANT reports at the end of the previous pass, and off after the pass. The board draws nothing between passes and needs no wake circuit; PMIC_EN goes only to the maintenance header J2.

Derating (NASA EEE-INST-002)

Part type	Rule applied	How the board meets it
Ceramic capacitors	Voltage ≤ 60 % of rated	Bus side: 100 V parts at 36 V (36 %). 5 V rails: 25 V parts (21 %). Logic and analog rails: 25 V or 100 V parts.
Ceramic capacitors on rails below 10 V	Rated ≥ 100 V, or flight lots screened to MIL-PRF-123 Group B (low-voltage humidity)	Values up to 100 nF are 100 V rated (0805). Larger values (1 to 22 µF, 25 V) need Group B screened lots; this goes in the procurement specification.
Tantalum, aluminum electrolytic	Tantalum ≤ 0.5 × rated with series resistance; aluminum electrolytic not listed	None on the board.
Chip resistors	Power ≤ 0.5 × rated (the standard allows 0.6), voltage ≤ 0.8 × rated; linear derating from 70 °C to zero at 118 °C	All 0805 resistors dissipate under 10 mW except the 120 Ω terminations, which are 1206 (0.10 W of 0.25 W, 41 %).
Metal-strip shunts	Power ≤ 0.5 × rated	R17, R18 (20 mΩ, 1206, 0.25 W): 94 mW at SW1's highest limit, 38 %.
Fuse (2 to 15 A)	Current ≤ 50 % of rating, less 0.2 %/°C above 25 °C	F1 4 A: 1.72 A allowed at 60 °C, against 1.25 A maximum input.
Schottky diodes	PIV ≤ 0.7 × rated, forward current ≤ 0.5 × rated	D4: 5 V of 70 V; 6.1 mA of 70 mA.
Connectors	Contact current derated to the wire gauge (26 AWG: 1.4 A)	J1: 0.42 A per power contact.
ICs	Supply within the recommended range	TPS26636 at 36 V against a 60 V recommended maximum (60 %).

Default case sizes: resistors 0805 (0.125 W), capacitors 0805 unless the value names a size.

Commercial parts fitted where the space-grade part cannot be bought

Ref	Fitted	Space-grade variant
U1	TPS26636PWPR	ZES723LCL
U5, U6	TPS259470LRPWR	TPS7H2201-SEP
U7, U8	INA293B2IDBVR	INA951-SEP
U18	TPS22919DCKR	TPS7H2221-SEP
U19	REF5040IDR	REF5040MDREP (enhanced product)
U10, U11, U13, U14, U15, U16, U17	TPS73801DCQR, TL7700CPWR, MCP2518FDT-E/SL, SN65HVD233DR, SN65C1168EPWR (two), ADC128S102CIMTX/NOPB	SEP versions where TI makes one, listed in the parts list
M1	Raspberry Pi CM5004016 (4 GB, 16 GB eMMC, no wireless)	none

The space-grade parts kept are fuse F1 (AEM P700L-72-4.0) and the four TE 44907 thermistors (GSFC 311P18-07); the commercial fallback for the thermistors is TE 44006RC.

CM5 pin allocation

All GND pins go to GND, the six 5V pins to 5V_CM5, and CM5_3.3V (pins 84 and 86) to CM5_3V3. Every other pin is unconnected.

Pin	CM5 name	Net
20	EEPROM_nWP	EEPROM_NWP (10k to GND: boot EEPROM write-protected)
24	GPIO26	HEAD_TRIG (use to be set with XIMEA)
28	GPIO13 (UART4 RX)	HOST_BULK_RXD
29	GPIO16	SW2_EN
30	GPIO6	WDI
31	GPIO12 (UART4 TX)	HOST_BULK_TXD
34	GPIO5 (UART2 RX)	HOST_CMD_RXD
37	GPIO7 (SPI0 CE1)	ADC_NCS
38	GPIO11 (SPI0 SCLK)	SPI0_SCLK
39	GPIO8 (SPI0 CE0)	CAN_NCS
40	GPIO9 (SPI0 MISO)	SPI0_MISO
41	GPIO25	CAN_NINT
44	GPIO10 (SPI0 MOSI)	SPI0_MOSI
46, 47	GPIO22, GPIO23	not connected
48	GPIO27	HOST_BULK_DE
49	GPIO18	PPS_IN
50	GPIO17	HOST_TLM_DE
51	GPIO15 (UART0 RX)	MAINT_UART_RX
54	GPIO4 (UART2 TX)	HOST_TLM_TXD
55	GPIO14 (UART0 TX)	MAINT_UART_TX
78	GPIO_VREF	CM5_3V3 (3.3 V GPIO)
92	PWR_Button	PWR_BUT (to J2)
93	nRPIBOOT	NRPIBOOT (to J2)
99	PMIC_Enable	PMIC_EN (to J2)
103, 105	USB_N, USB_P	MAINT_USB_N/P (to J2)
128, 130	USB3-0-RX_N/P	HEAD_SSRX_N/P
134, 136	USB3-0-DP/DM	HEAD_USB2_P/N
140, 142	USB3-0-TX_N/P	HEAD_SSTX_N/P (AC capacitors are on the CM5)

Connector pin maps

J1, harness header (Harwin Gecko G125-MV12605L1P, 2 × 13, 1.25 mm). Wired to the Glenair 801 power receptacle and the 15-way micro-D. Mate: housing G125-2042696L0 with crimp contacts G125-0010005 (power) and G125-0020005 (signal). Harwin numbers the pins row by row (1 to 13, then 14 to 26, with 14 across from 1); each pair sits across the two rows, and the pins fall into the three isolation domains.

Column	Row 1 pin	Net	Row 2 pin (across)	Net	Domain
1-3	1, 2, 3	BUS_IN (9 to 36 V)	14, 15, 16	PRI_RTN	host side
4	4	HOST_THERM_A (RT4)	17	HOST_THERM_B (RT4)	host side
5	5	not connected	18	not connected	gap
6	6	CMD_IN_P	19	CMD_IN_N	payload
7	7	TLM_P	20	TLM_N	payload
8	8	BULKOUT_P	21	BULKOUT_N	payload
9	9	BULKIN_P	22	BULKIN_N	payload
10	10	PPS_IN_P	23	PPS_IN_N	payload
11	11	CANH	24	CANL	payload
12	12	GND (CAN ground and signal reference)	25	not connected	payload
13	13	not connected	26	CHASSIS	chassis

J2, maintenance header (Molex 87831-1420).

Pin	Net	Pin	Net
1, 2, 5, 8, 14	GND	9	NRPIBOOT
3, 4	MAINT_USB_P, MAINT_USB_N	10	PMIC_EN
6	MAINT_UART_TX	11	PWR_BUT
7	MAINT_UART_RX	12	EEPROM_NWP
		13	MAINT_3V3_REF (CM5_3V3 through 100 Ω)

J3, head link (16-way, connector to be set with XIMEA).

Pins	Net
1 to 3	5V_HEAD
4, 7, 10, 13, 15, 16	GND
5, 6	HEAD_SSTX_P/N
8, 9	HEAD_SSRX_P/N

Pins	Net
11, 12	HEAD_USB2_P/N
14	HEAD_TRIG

Before the flight board

- Finish the 22 remaining connections and rerun the design-rule and barrier checks.
- Confirm J1's 26-way numbering and its latch retention on a 2.4 mm board, and the land patterns not yet checked against vendor drawings (J2 numbering beyond circuits 1 and 2, TPD4E05U06, ABM8AIG, Keystone 1593-2 turrets, XGL5050, the CM5 pad mask).
- Choose the bus TVS D5 against the host's surge specification, and fit the chassis tie R14/C19 to the host's grounding plan.
- On the bench: check the damped input filter against the converter's input impedance and run a Class A emissions test; measure the CM5 boot time (under 57 s for the watchdog), the boot-peak current and the head current, then set the switch limits and software thresholds.
- With XIMEA: the head current at 0.5 frame/s, the flex connector J3 and the use of the trigger line.
- Procurement: Group B low-voltage humidity screening (MIL-PRF-123) for the 1 to 22 µF ceramics on rails below 10 V, and radiation screening or lot testing for the commercial ICs, the ESD2CAN24-Q1 and the TPD4E05U06.
- From the fabricator: the laminate's dielectric constant and loss tangent and the impedance-controlled widths.

Sources

Pin tables and equations come from the vendors' own documents:

- CM5 datasheet RP-008180-DS-7, Tables 1 and 4
- TI datasheets:
  - TPS2663 (TPS26636), TPS25947, INA293, TPS22919, REF50xx
  - TPS73801, TL7700, TPS3435
  - ADC128S102, SN65HVD233, SN65C1168E
  - ESD2CAN24-Q1, TPD4E05U06
  - Space-grade variants named in MPN_ALT: TPS7H2201-SEP, INA951-SEP, TPS7H2221-SEP, TL7700-SEP
- Nexperia BAS70W-Q datasheet: pin 1 anode, pin 2 not connected, pin 3 cathode; 70 V, 70 mA
- Harwin G125-MV12605L1P: 26 contacts, 2.8 A per contact (2 A simultaneous at 25 °C), −65 to +150 °C, latch barbs for 1.6 mm boards
- Abracon ABM8AIG-40.000MHZ-12-2Z-T3: CL 12 pF, ±20 ppm, ±50 ppm, −40 to 125 °C, ESR 35 Ω

- TDK-Lambda CCG instruction manual C286-04-11E, Table 6-2 (trim-up equation, read from the rendered page)
- TDK-Lambda CCG10-24 specification
- TDK CCG6/CCG10 quick manual section 8, filter topology and pin positions (read from the rendered page)
- Molex 87831-1420
- MCP2518FD: Microchip's site refuses automated access. Its pinout was read from Microchip's own datasheet DS20006027B, in a copy hosted by SparkFun.
- J. Olson, "Determining current carrying capacity", PCD&F, 2009: IPC-2152 vacuum behaviour (conductors 35 to 55 % hotter than in air), read 11 Oct 2026.